

Frontier Atlas Architecture

IMPLEMENTED NOW

Bounded asyncio workers use a queue, concurrency limits, timeouts, exponential backoff with jitter, HTTP 429 handling, and Retry-After. A 413 provider response reduces LLM chunk size. Gemini Flash, Groq Llama, and DeepSeek adapters use environment keys, isolate failures, validate JSON with Pydantic, and fall back in order. Deterministic source parsers preserve URLs, dates, raw names, and collection timestamps. SQLite unique keys provide persistent idempotency; CSV/JSONL and Sheets dry-run exports are local outputs.

PRODUCTION SCALE DESIGN

For 500k+ records, adapters publish to a durable queue with source-specific rate limits and backpressure. Stateless async workers store raw responses in object storage, then write canonical validated records to PostgreSQL with unique URL/content keys. Redis supplies distributed locks, quotas, quarantine state, and coordination. Horizontal workers consume bounded batches; retry queues and dead-letter storage preserve failure recovery context. PostgreSQL, Redis, object storage, Kubernetes, vector storage, and graph storage are design decisions, not local deployments.

Reliability and data integrity

Freshness tracking stores the source date, date method, collection timestamp, and rejection reason. Absolute dates, ISO timestamps, meta tags, time tags, JSON-LD, RSS dates, relative dates, and yesterday are handled conservatively; unverifiable dates are rejected for fresh feeds. Entity resolution normalizes legal suffixes and aliases, uses exact matching first, and records fuzzy raw name, canonical name, method, and confidence. Short names do not receive aggressive fuzzy matches. Missing facts remain null. GitHub repositories are accepted only when explicitly supported by the source, and stars come only from the GitHub API.

Operational controls

Observability records source, URL, status, latency, retries, provider, freshness, and extraction result without secrets. A provider with malformed JSON, invalid schema output, 429, timeout, 413, or terminal failure is isolated and the next provider is attempted. Idempotent writes make at-least-once retries safe; this is not a mathematical exactly-once guarantee. Vector projections support semantic retrieval. Graph projections model companies, products, papers, authors, and repositories but never replace canonical storage.

Responsible acquisition

Official APIs are preferred, followed by RSS/Atom feeds and permitted static HTML. Source adapters are configurable and blocked sources are logged, backed off, quarantined, and continued past. The pipeline does not solve CAPTCHAs, bypass authentication, evade Cloudflare or Datadome controls, scrape aggressively, or use paywalled content without permission. Blocked sources must be escalated to an approved API, licensed provider, or partnership.

Local verification status

The deterministic suite passes 17 tests. Live source counts vary with network access, freshness, and source availability. Provider adapters are implemented and require GEMINI_API_KEY, GROQ_API_KEY, or DEEPSEEK_API_KEY for live execution. Google Sheets has a local dry-run transformation and requires user-supplied credentials for live export.